CSC1180 Computing for Business Team Project (12202)

Mid-Term Deliverable

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1. Initial Business Model Summary

1.1. Project Concept and Overview

LevelUp! is a web-based marketplace application designed to improve the way private lessons are scheduled and booked in Ireland. The platform supports a range of lesson types, including academic grinds, sports coaching, and music lessons. Instructors are able to advertise their services and define available time slots in advance, while students, or parents booking on their behalf, can browse offerings and instantly book sessions that suit their schedules. By removing the need for prolonged back and forth communication through messaging platforms such as WhatsApp or email, LevelUp! aims to significantly reduce scheduling inefficiencies and lost booking opportunities.

The core concept of LevelUp! is centred on scheduling efficiency rather than direct communication. In the current private lessons market, instructors and learners frequently rely on informal messaging to arrange sessions. When availability does not align immediately, communication often breaks down, leading to missed opportunities and frustration on both sides. LevelUp! addresses this problem by introducing a booking-first model in which availability is clearly presented and sessions are secured instantly, creating a more reliable and predictable booking experience across multiple lesson categories.

1.2. Problem Statement

Private lessons in Ireland remain largely organised through informal communication channels such as WhatsApp and text messaging. Primary research conducted with students and parents showed that most lesson arrangements involve repeated back-and-forth messaging, lost conversations, unclear availability, and inconsistent pricing information. Parents also expressed concerns regarding safety, lack of verified instructor credentials, and the absence of a central platform to compare tutors. Tutors and coaches reported spending significant time coordinating schedules, chasing payments, and handling cancellations, creating administrative burden and income uncertainty. These findings highlight the need for a structured, transparent, and trusted booking system to replace existing informal scheduling practices.

These challenges become more apparent during periods of increased demand, such as examination preparation, sports competition seasons, and music performance rehearsals. Delayed responses or scheduling conflicts regularly result in abandoned bookings. The absence of a structured booking mechanism creates inefficiencies that negatively affect both learners and instructors. LevelUp! seeks to address this gap by providing a centralised and transparent platform for managing lesson availability and bookings.

1.3. Value Propositions

LevelUp! delivers value by simplifying and standardising the booking process for private lessons. For students and parents, the platform provides a clear and immediate view of instructor

availability, allowing bookings to be made quickly without uncertainty. This improves access to high-demand instructors and reduces the time required to secure lessons.

For instructors, including teachers, sports coaches, and music tutors, LevelUp! reduces the administrative overhead associated with managing bookings. By allowing instructors to define availability in advance, the platform offers greater control over scheduling and promotes efficient time management. The result is a structured and predictable workflow that benefits both sides of the marketplace.

1.4. Target Users and Customers

The target users of LevelUp! include secondary school students preparing for Junior Cycle and Leaving Certificate examinations, and individuals pursuing private sports coaching or music lessons. In many cases, parents or guardians act as the primary decision-makers when booking, particularly for younger students. On the supply side, the platform targets independent instructors operating across Ireland, including academic tutors, sports coaches, and music teachers.

The system supports both in-person and online lessons, enabling adoption across urban and suburban regions. By accommodating multiple lesson types and user groups, LevelUp! aims to provide a flexible and inclusive booking solution within the Irish private lessons market.

1.5. Market Opportunity and Size

Ireland has a well-established culture of private tuition and extracurricular coaching, with academic grinds forming a significant part of the education landscape. Research indicates that more than half of Leaving Certificate students, predominantly from higher-income households, avail of grinds to improve their chances of accessing third-level education [1]. In 2025, approximately 65,000 students sat the Leaving Certificate examinations [2], representing a substantial and recurring annual cohort of potential users for platforms supporting private tuition. This market renews itself each academic year as similarly sized student groups progress through the education system, creating consistent demand for private academic support. In addition to this core segment, private sports coaching and music lessons are widely used across Ireland for skill development, competitive preparation, and personal enrichment, further expanding the potential user base.

While there is no single official figure quantifying the entire Irish private lesson industry, it operates as part of the wider private tuition ("grinds") sector, which is estimated to be worth approximately €30–€80 million per year in Ireland [3]. A Total Addressable Market (TAM) includes all private tutoring, sports coaching, and music lesson activity. The Serviceable Available Market (SAM), representing services accessible through online or hybrid delivery, forms a subset of this sector. A conservative Serviceable Obtainable Market (SOM), estimated at 2–3% of the Serviceable Available Market, represents a realistic early-stage opportunity for a new digital booking platform, based on conservative market entry assumptions.

Primary research conducted with teachers revealed that only 12% currently offer private grinds, while 64% expressed interest in providing lessons in the future. The most commonly reported barriers were lack of client connections, excessive administrative communication, uncertainty about how to begin offering grinds, and the time required to advertise services. This indicates a strong latent supply of instructors who would engage with a platform that reduces administrative workload and improves visibility, supported by primary research findings presented in Appendix F [5].

Taken together, these findings confirm a substantial market opportunity on both the demand and supply sides. Rather than competing directly with broad tutor discovery platforms, LevelUp! targets a specific operational gap shared across academic, sports, and music instruction, namely the inefficiency of scheduling and booking. By focusing on this cross-disciplinary problem, the platform maintains a clear and scalable value proposition. Even limited adoption within these markets represents a meaningful opportunity for growth.

1.6. Revenue streams

Several potential revenue models have been identified to support the long-term sustainability of the LevelUp! platform. A commission-based fee may be applied to completed bookings made by students or parents. Instructors may also opt for subscription plans that provide enhanced scheduling tools or promotional features. In addition, the platform may offer paid advertisement and featured listing options, allowing instructors or coaching services to increase their visibility during periods of high demand. Together, these revenue streams provide a realistic commercial pathway while aligning with the platform's core booking and scheduling functionality.

Table 1.

Revenue	Description	Target User
Commission Fees	Percentage applied to completed lesson bookings	Students/parents
Instructor Subscriptions	Monthly subscription for enhanced scheduling and promotional tools	Instructors
Featured Listings	Paid option to increase visibility in search results	Instructors
Advertisement Revenue	Google Ads	External

2. Summary Functional Specification & Technical Description

2.1. System Overview

LevelUp! is a Django-based web application that provides a centralised platform for booking private lessons in Ireland, including academic grinds, sports coaching, and music lessons. The system is designed around two main types of users: instructors, who offer lessons and manage their availability, and students, or parents booking on their behalf, who search for lessons and make bookings. The primary goal of the system is to remove informal scheduling and replace it with a clear, structured booking process.

The application follows Django's Model–Template–View architecture. Data such as users, lesson information, availability slots, and bookings are stored in a relational database. Business logic is handled by Django views, while templates are used to render the user interface. This structure allows the system to remain organised and easy to extend as new features are added.

2.2. Core System Functionality

Users register for an account and are assigned a role as either an instructor or a student/parent. Instructors create profiles describing the lessons they offer, including academic subjects, sports coaching services, or music tuition. Each instructor can create lesson listings and define their availability by generating individual time slots that represent specific dates and times when lessons can be booked.

Students and parents can browse instructor listings and view available time slots without the need for direct communication with instructors. When a suitable slot is selected, a booking request is created and confirmed instantly. Once confirmed, the slot becomes unavailable to other users, preventing double booking. Both instructors and students can view and manage their upcoming and past bookings through dedicated dashboard pages.

System administration is handled through Django's built-in admin interface, which allows oversight of users, listings, availability slots, and bookings during the prototype stage.

To ensure correct system behaviour, the following core rules apply:

- Only instructors can create listings and availability slots.
- A time slot must have a valid start and end time, with the start time preceding the end time.
- Availability slots belonging to the same instructor cannot overlap.
- A slot can only be booked once and must immediately change status upon confirmation.
- Only authenticated users can create or manage bookings.

2.3. Non-Functional Requirements

In addition to core functionality, the system must meet several non-functional requirements. User data must be stored securely, and access to the platform must be restricted to authenticated users. As the application operates within the European Union, data handling must align with GDPR principles, including minimising the collection of personal data.

Instructor identity verification is required to ensure legitimacy and build trust. Payment processing will be handled through trusted third-party providers such as Stripe or PayPal to ensure PCI-DSS compliance, fraud protection, and to avoid storing sensitive card data directly on the platform.

The system must also be responsive and reliable, particularly during periods of high demand such as exam preparation or competitive sports seasons. The user interface should be intuitive and easy to navigate, allowing users to complete bookings with minimal effort. While the

prototype is not designed for large-scale commercial deployment, the underlying architecture should support future growth and feature expansion.

2.4. Technical Implementation

The prototype is implemented using the Django web framework, which provides built-in tools for user authentication, database management, and administrative control. Django's Object-Relational Mapping layer is used to define and manage the core data models, including users, instructor profiles, availability slots, and bookings.

During development, a lightweight database is used to simplify setup and testing. The system is structured so that it could later be deployed using a more scalable database solution if required. Static assets such as stylesheets and images are managed using Django's static files system. This technology stack allows the project to focus on delivering core functionality while following established software development practices.

2.5. Development Challenges

One of the main technical challenges in developing LevelUp! is ensuring that booking operations are handled correctly when multiple users attempt to book the same time slot. The system must ensure that a slot can only be booked by one user and that availability is updated immediately after a booking is confirmed.

Additional challenges include managing date and time information accurately, enforcing role-based access across the platform, and ensuring that the system remains simple to use despite supporting multiple lesson types. These challenges are addressed through careful model design and validation within the Django framework.

Beyond technical implementation, the system must address trust and safety requirements identified in user research. Parents require assurance that instructors are verified and that communication occurs within a monitored platform rather than personal messaging applications. Tutors seek reduced administrative effort and guaranteed payment handling. Designing booking workflows that remain simple while incorporating verification, parental oversight, and GDPR-compliant data handling presents a key usability and technical challenge.

2.6. Prototype Scope

The prototype developed for this project focuses on demonstrating the core concept of structured lesson booking. It includes user registration and authentication, instructor profile creation, availability management, and booking functionality. Administrative oversight is provided through Django's admin interface.

Features such as payment processing, in-app messaging, and advanced reporting are not included in the prototype. These features are considered outside the scope of the current project and are identified as potential areas for future development.

3. Project Timeline

3.1. Project Plan Overview

The LevelUp! project is developed following a staged approach, progressing from requirements definition and system design through to implementation, testing, and final documentation. The timeline is designed to ensure that the core booking functionality is implemented early, allowing sufficient time for refinement, testing, and preparation of final project deliverables. Regular internal milestones are used to track progress and ensure that the project remains aligned with academic deadlines.

The work is divided among team members according to individual strengths. Responsibility for backend development, including database design and booking logic, is assigned to team members with experience in server-side programming. Frontend development and user interface design are assigned to team members focusing on usability and presentation. Documentation, testing, and report preparation are shared across the group to ensure consistency and quality in written deliverables.

3.2. Development Phases

The project begins with finalising requirements and producing system designs, including data models and user flows. Development proceeds with Django project setup and database model implementation. Instructor features, including profile and availability management, are developed first, followed by student-facing booking features. Integration, validation, and testing are then performed to ensure correct system behaviour. The final phase focuses on documentation, bug fixing, and preparation for submission and project presentation.

3.3. Gantt Chart & Task Allocation

Figure 1 presents the project schedule in the form of a Gantt chart. The chart outlines the progression of the project from proposal development through system implementation, testing, and final delivery, aligned with module submission deadlines. Each task is assigned to specific team members, with hatch patterns used to distinguish individual and shared responsibilities. This provides clear visibility of workload distribution and ensures accountability throughout the project lifecycle.
(Figure 1.)

Task Allocations

Table 2.

Task	Description	Group Member
Project Proposal & Advisor Coordination	Define project idea, form team, secure advisor, submit proposal	Dylan & James
Requirements & System Design	Define functional requirements, system architecture, ERD, and user flows	Dylan & James
Django Project Setup	Create Django project, configure environment, set up database	Dylan & James
Database Models	Implement User, Instructor, Listing, Availability Slots, and Booking models	Dylan & James
Instructor Features	Develop instructor profile management and availability slot creation	Dylan

Task	Description	Group Member
Student Features	Develop student browsing, booking workflow, and dashboards	James
User Interface Development	Implement Django templates and frontend layout	James
Mid-Term Documentation	Prepare business model, functional spec, and timeline report	Dylan & James
System Integration	Connect frontend templates with backend logic	Dylan & James
Testing & Debugging	Test booking logic, validation, and fix defects	Dylan & James
Final Documentation & Presentation	Prepare final report, demo material, and presentation	Dylan & James

3.4. Risk Management

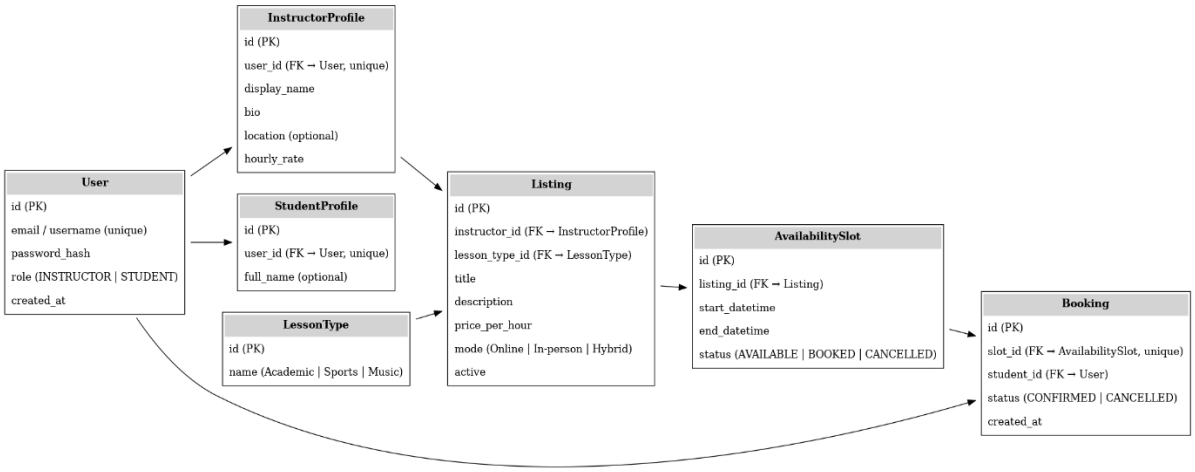
The development of the LevelUp! platform involves technical, project, and operational risks that may impact successful delivery. Identifying these risks early allows appropriate mitigation strategies to be put in place. The most significant risks relate to implementation complexity, time constraints, and user adoption challenges. Table 3 summarises key project risks, their potential impact, and proposed mitigation measures.

Table 3.

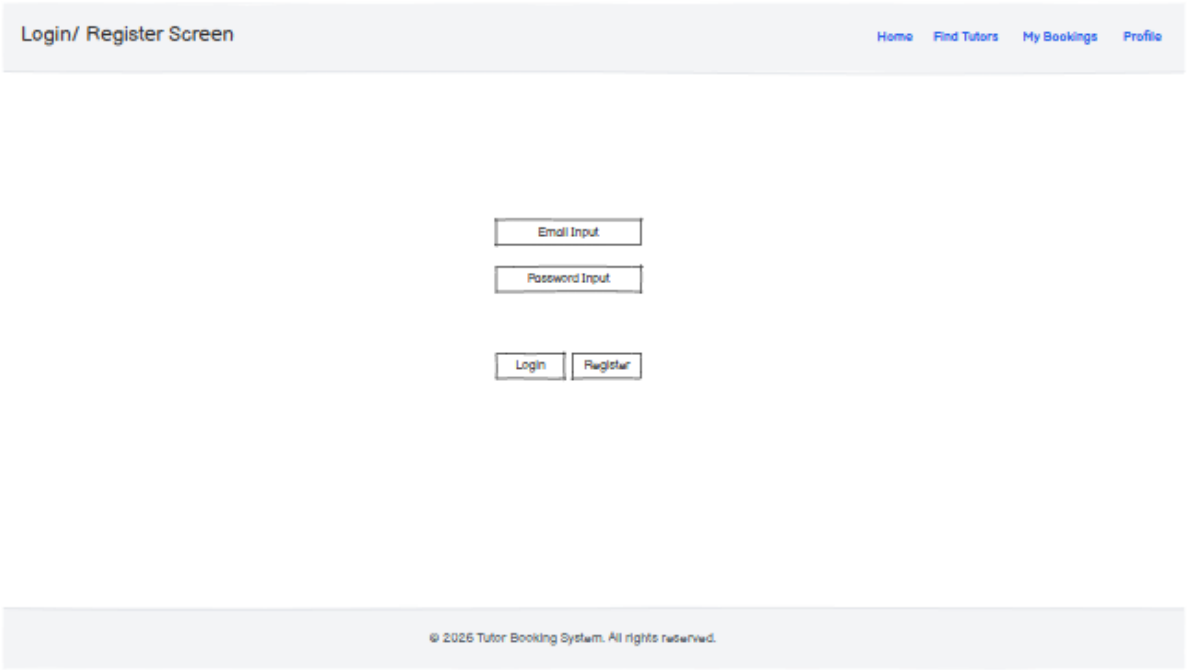
Risk	Potential Impact	Mitigation Strategy
Double booking logic not functioning correctly	Conflicting bookings and loss of user trust	Implement database constraints and test concurrency handling early
Time constraints due to academic workload	Delayed development and incomplete prototype	Maintain weekly task planning and buffer time for testing
Difficulty integrating frontend and backend	System instability or incomplete user workflows	Develop features incrementally and conduct early integration testing
Limited user testing opportunities	Reduced feedback on usability	Conduct informal testing with classmates and instructors
Low initial user adoption	Platform not validated with real users	Focus prototype on core value proposition and clear demonstration

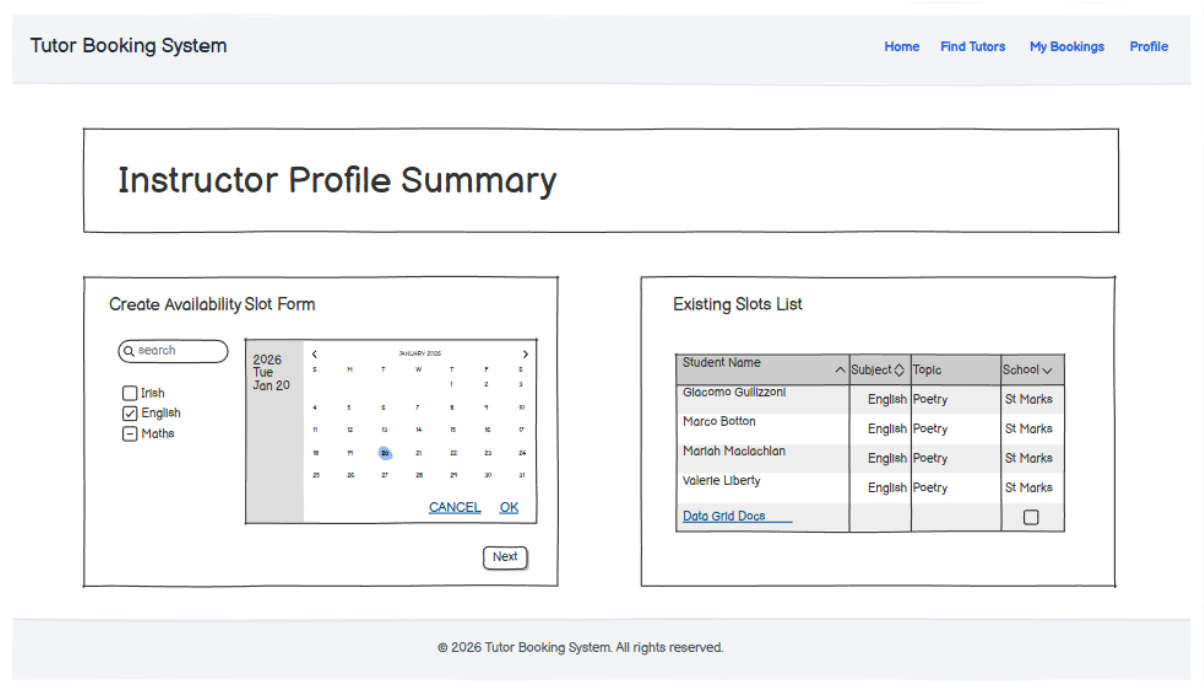
4. Appendices

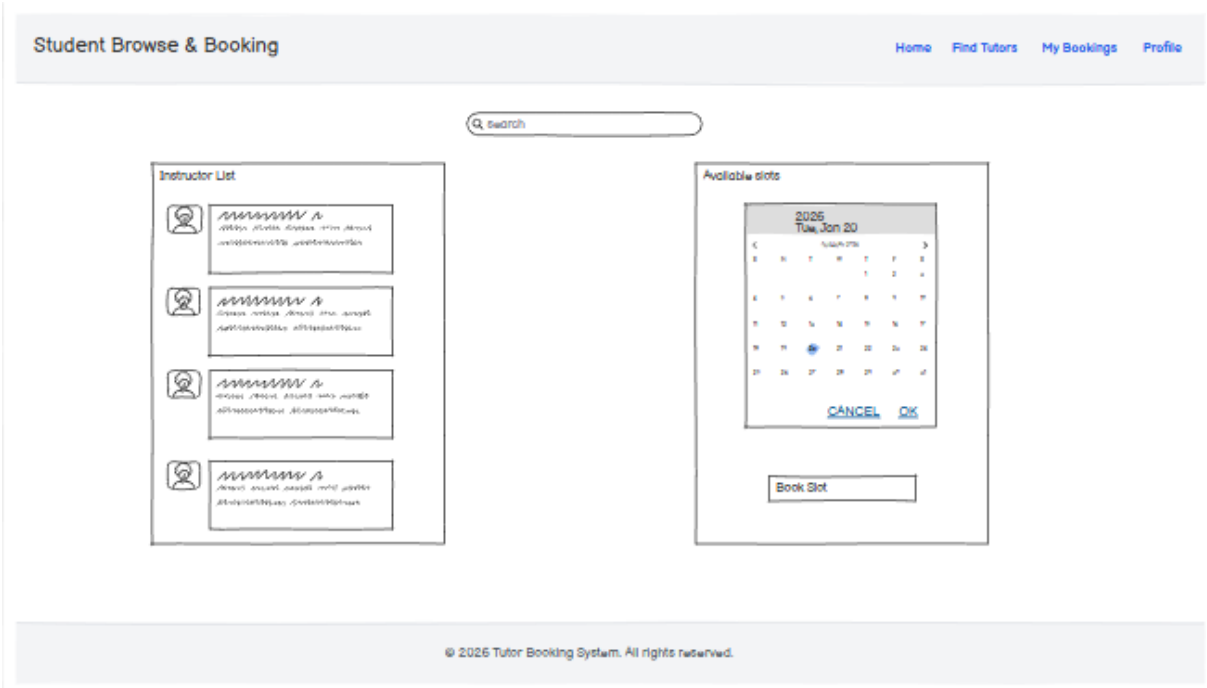
Appendix A – ER Diagram



Appendix B – User interface Wireframes [6]






Student Browse & Booking
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Find Tutors
My Bookings
Profile
Search
Instructor List
Available slots
2026
Tue, Jan 20
CANCEL
OK
Book Slot
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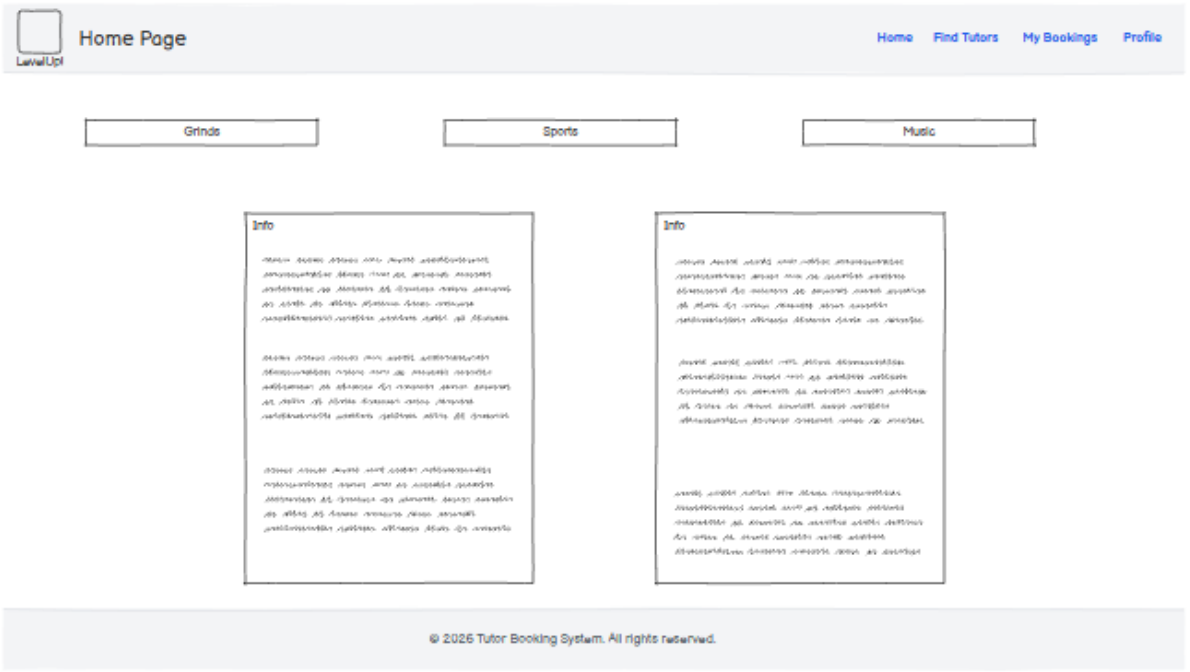
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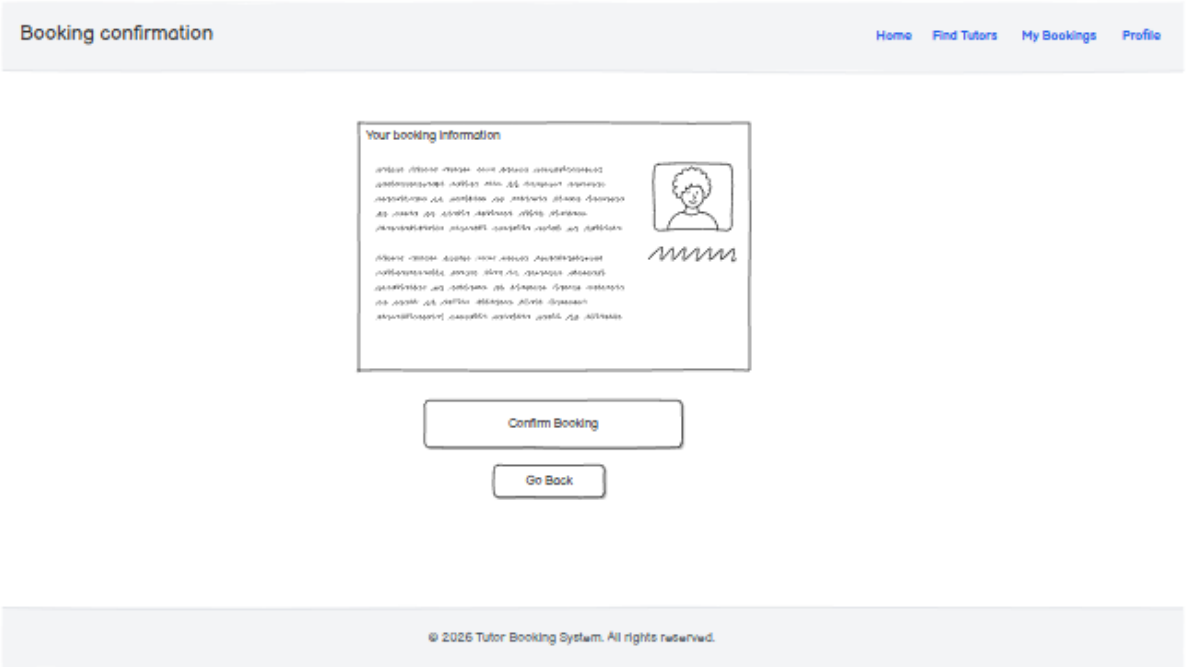
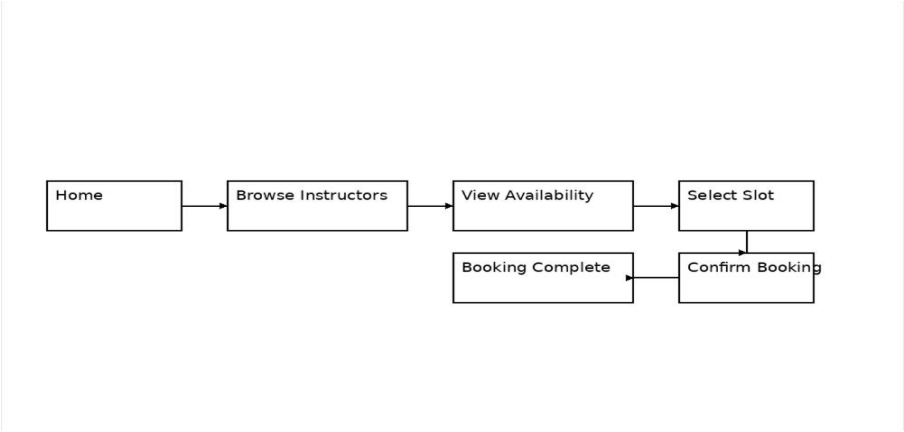
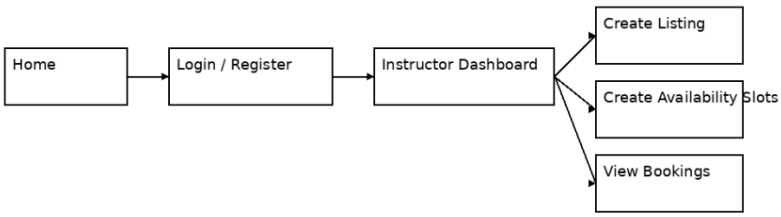


Appendix C – User Flow Diagram





Appendix D – Figure 1. Gantt Chart

Appendix E – Extended Functional Requirements

User Registration and Authentication

The system must allow users to register using a valid email address and password. Users must select a role (instructor or student/parent) during registration. Duplicate email registrations must be prevented. Secure login and logout functionality must be provided.

Instructor Profile Management

Instructors must be able to create and edit a profile including lesson description, subject or discipline, and pricing information. Listings may be activated or deactivated as required.

Availability Slot Management

Instructors must be able to create availability slots by selecting dates and times. Overlapping availability slots for the same instructor must be prevented. Un-booked future slots may be deleted.

Lesson Browsing

Students and parents must be able to browse instructor listings and filter results by lesson type or subject. Only active listings are visible.

Booking Creation

Students and parents must be able to select available time slots and create bookings. The system must verify slot availability before confirming. Confirmed bookings must immediately update slot status to prevent double booking.

Booking Management

Students and instructors must be able to view upcoming and past bookings. Instructors must be able to cancel bookings where necessary.

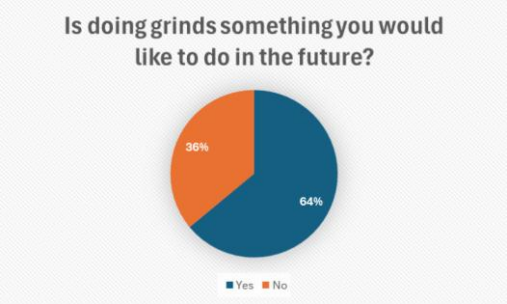
Administrative Oversight

Administrators must be able to manage users, listings, availability slots, and bookings through the Django admin interface.

Data Validation

All date and time inputs must be validated. Clear error messages must be displayed where invalid input is detected.

Appendix F – Primary Research Graphs [5]



5. References

[1] - Irish Times: https://www.irishtimes.com/ireland/education/2024/09/12/most-leaving-cert-students-availing-of-grinds-but-little-evidence-of-any-gain-for-higher-achieving-students/#:~:text=Some%2055%20per%20cent%20of,Ireland%20(55%20per%20cent).

[2] - RTE: https://www.rte.ie/news/2025/0822/1529612-leaving-cert/#:~:text=Almost%20record%20numbers%20sitting%20exams,Read%20the%20full%20story%20here.

[3] – The Irish Times: https://www.irishtimes.com/ireland/education/2024/10/20/were-fully-booked-how-grind-schools-became-big-business/?fbclid=IwY2xjawPcZdRleHRuA2FlbQIxMQBzcnRjBmFwcF9pZBAyMjIwMzkxNzg4MjAwODkyAAEeLGF8DCOwp4XuYIBEvaB8w6FVkzoDgF1mudh15QUm5YYetZ2iaHOopL_xuz4_aem_a8FkBLnLW6-oqc1Okc-7Ng/

[4] - Gantt Chart: https://www.canva.com/design/DAG-5W13imk/ds-pkKGyEY6yRjTcgTjCMg/edit?utm_content=DAG-5W13imk&utm_campaign=designshare&utm_medium=link2&utm_source=sharebutton

[5]Primary Research Survey (2025). Conducted among 25 teachers to analyse current grind participation and barriers to entry. (Appendix F)

[6] – Balsamiq – Wireframes : https://balsamiq.com/?gad_source=1&gad_campaignid=203404003&gbraid=0AAAAAD3BuzNKjuYIAJwK6oO9zTKvs7LLa&gclid=CjwKCAiA7LzLBhAgEiwAjMWzCHhLHSYcKallNS_AkjMbMjQoEg0w8eFT4F8RBoTCE7cVQYbfMg0NqxoC7oEQAvD_BwE